PRACTICAL LABORATORY ASSIGNMENT

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Course	GRC 103 Week 2 Lab A

Phase 1: Basic Metric Calculations

Step 1: Calculate Fundamental Rates

Campaign	Email Open Rate	Click-Through Rate	Credential Submission Rate	Data Entry Rate
A	(82/100)×100 = **82%**	(47/100)×100 = 47**%**	(18/100)×100 = 18**%**	(5/100)×100 = 5**%**
B	(200/250)×100 = 80**%**	(150/250)×100 = 60**%**	(75/250)×100 = 30**%**	(25/250)×100 = 10**%**
C	(72/80)×100 = 90**%**	(60/80)×100 = 75**%**	(12/80)×100 = 15**%**	(8/80)×100 = 10**%**

Step 2: Calculate Efficiency Metrics

Conversion Rate from Open to Click:

- Campaign A: (47/82)×100 = **57.3%**
- Campaign B: (150/200)×100 = **75%**
- Campaign C: (60/72)×100 = **83.3%**

Post-Click Credential Submission Rate:

- Campaign A: (18/47)×100 = **38.3%**
- Campaign B: (75/150)×100 = **50%**
- Campaign C: (12/60)×100 = **20%**

Phase 2: Advanced Risk Quantification

Step 3: Calculate Organizational Risk Exposure

For Campaign B (most successful):

- Expected Credentials Stolen: 2,000 × (75/250) = **600 employees**
- Expected Data Breaches: 600 × 25% probability = **150 breaches**

Phase 3: Comprehensive Risk Assessment

Step 6: Complete Risk Assessment Matrix

Campaign	Overall Risk Score	Financial Exposure	Priority Level	Recommended Action
A	MEDIUM-HIGH	$391,500,000	2	Enhanced Training
B	**CRITICAL**	**$652,500,000**	**1**	**IMMEDIATE MFA**
C	HIGH	$326,250,000	2	Executive Training

Step 7: Confidence Interval for Campaign A

Formula: $p \pm 1.96 \times \sqrt{[p(1-p)/n]}$

Working:

1. $p = 0.18$, $n = 100$
2. $p(1-p) = 0.18 \times 0.82 = 0.1476$
3. $0.1476 \div 100 = 0.001476$
4. $\sqrt{0.001476} = 0.0384$
5. Margin of error = 1.96×0.0384 = **0.0753 (about 7.5 percentage points)**
6. Range = 0.18 ± 0.0753 = **10.5% to 25.5%**

Phase 4: Executive Reporting

Step 8: Comprehensive Executive Briefing

To: Board of Directors, CISO, CFO
From: GRC Risk Analysis Team
Date: October 31, 2026
Subject: CRITICAL: Quantitative Phishing Risk Assessment and $652.5M Modeled Exposure

1. Executive Summary

Three phishing simulations covering 430 employees produced 105 credential submissions, a pooled rate of 24.4%. The “IT System Password Reset” campaign was the most successful, with a 30% credential rate. Projected across 2,000 employees, that is 600 compromised accounts. At a 25% probability that stolen credentials lead to a major breach, this means 150 breaches per year and a modeled Annualized Loss Expectancy of $652.5 million. A $490,000 control portfolio is expected to cut this exposure by 90%, to about $65.25 million. Immediate action is recommended.

2. Key Quantitative Findings

Risk Metric	Value	Industry Average*	Severity
Overall credential theft rate (pooled)	24.4%	15.3%	HIGH
Maximum campaign success rate (B)	30%	18.7%	CRITICAL
Annualized Loss Expectancy	$652.5M	$285M	SEVERE
MFA implementation ROI	724,175%	350%	EXCELLENT

3. Detailed Financial Analysis

Current state risk (modeled worst case, based on Campaign B):

- Annualized Loss Expectancy: **$652,500,000**
- Expected major breaches per year: **150**
- Per-employee risk: $652,500,000 ÷ 2,000 = **$326,250**

Proposed investment portfolio:

Control	Cost	Risk Reduction	Net Benefit	ROI	Priority
MFA implementation	$90,000	$651,847,500	$651,757,500	724,175%	1
Enhanced training	$100,000	$391,500,000	$391,400,000	391,400%	2
Email filtering	$50,000	$130,500,000	$130,450,000	260,900%	3

4. Strategic Recommendations

Immediate actions (Q1):

- Implement MFA enterprise-wide: $90,000
- Launch targeted phishing simulation program: $25,000
- Conduct emergency security awareness training: $100,000

Strategic initiatives (next fiscal year):

- Deploy advanced email security gateway: $50,000
- Establish continuous security monitoring: $150,000
- Implement security behavior analytics: $75,000

5. Expected Risk Posture After Controls

Metric	Current	With Controls	Reduction
Credential theft rate	24.4%	2.4%	90%
Annual Loss Expectancy	$652.5M	$65.25M	90%
Expected breaches per year	150	15	90%

Portfolio summary:

- Total investment required: **$490,000**
- Total risk reduction: $652,500,000 × 0.90 = **$587,250,000**
- Net financial benefit: $587,250,000 − $490,000 = **$586,760,000**
- Portfolio ROI: $586,760,000 ÷ $490,000 × 100 = **about 119,747%**

Break-even (bonus challenge): $490,000 ÷ $4,350,000 = **0.11 breaches**. The program pays for itself if it prevents roughly one-ninth of a single breach, against the 135 breaches (150 − 15) it is expected to prevent.